

GALA: ANATOMY-AWARE LANGUAGE-MOTION ALIGNMENT FOR EFFICIENT TEXT-TO-MOTION RECTIFIED FLOW

Anonymous submission

Author names and affiliations required before submission

ABSTRACT

Text-to-motion generation must couple language semantics with skeletal kinematics, yet most latent generators use pooled text conditioning or denoise raw poses at high cost. Existing methods lack an explicit bridge between skeletal topology, fine-grained semantics, and the generative trajectory. GALA establishes that bridge: a skeleton-graph VAE exposes five body-part latents; learnable anatomical queries align CLIP tokens to those parts; a rectified-flow DiT then transports noise under motion-space kinematic constraints, sampled with a 20–50 step Euler ODE. Under the official 20-replication protocol, GALA reaches R@3 0.806 / FID 0.272 at 20 NFEs and FID 0.224 at 50 NFEs on HumanML3D; on KIT-ML it obtains R@3 0.804 / FID 0.402. The graph tokenizer reconstructs at FID 0.003, while part indexing and kinematic supervision reduce 20-NFE generation FID from 0.337 to 0.272.

Index Terms— text-to-motion, rectified flow, skeleton graph, cross-modal alignment, human motion generation

1. INTRODUCTION

Text-to-motion (T2M) maps a sentence to a variable-length pose clip [1, 2]. HumanML3D [1] and KIT-ML [3] are the standard splits; the Guo protocol—R-Precision, FID [4], multimodal distance, Diversity, 20 replications—is the number reviewers compare [1, 2]. Diffusion on raw 263-D poses [2, 5] is faithful but slow; latent diffusion [6] and discrete tokens [7, 8, 9] push FID down at the cost of a codebook or a long denoising chain. Rectified flow [10, 11] replaces that chain with a direct ODE trajectory and enables low-NFE continuous generation.

A topology-aware variational tokenizer already yields a compact, near-lossless motion latent, so generation reduces to learning a *straight* path in that space that is explicitly aligned with language. Existing latent T2M methods lack an explicit structural bridge between skeletal topology, fine-grained language semantics, and the generative trajectory. GALA establishes this bridge by linking three stages that are not interchangeable: the graph tokenizer exists to expose anatomical parts; part-level alignment exists so language can address those parts; kinematic flow exists because reconstruction FID shows that fidelity is lost in the ODE, not in the VAE—skeleton graph $\rightarrow$ part latents $\leftrightarrow$ language tokens $\rightarrow$ kinematic rectified flow.

Contributions.

1. **Anatomy-aware motion tokenizer.** A CTR graph encoder plus stride-4 VAE maps 263-D poses to 256-D latents at 5 Hz, preserving skeletal dependencies and exposing part-structured features. Val reconstruction FID is 0.003.
2. **Part-indexed language-motion alignment.** Beyond pooled InfoNCE, five fixed-index queries condition the flow and are supervised by corresponding graph-pooled body parts; a collapse audit verifies that their attention profiles remain distinct.

3. **Kinematic-aware rectified flow.** An 8-block DiT learns a straight latent ODE under classifier-free guidance [12], with velocity, bone-length, and foot-contact penalties in decoded motion space, enabling 20–50 NFE sampling.

We compare against native-HumanML3D generators (MDM, MLD, M2DM, T2M-GPT, EMDM, MoMask) and treat GENMO [13] as one generalist SMPL baseline, not as the target.

2. RELATED WORK

Pose-space DDPMs [2, 5, 14] are faithful but slow. MLD [6] diffuses in a VAE latent—the same split we use, without a skeleton graph or a straight flow. T2M-GPT [7], M2DM [9], MoMask [8], and MMM [15] quantize motion; we stay continuous because a 256-D VAE already reconstructs at FID 0.003. Consistency models [16] and EMDM [17] cut the denoising chain; rectified flow [10, 11] learns a direct path that a DiT [18] can parameterize under CFG [12]. Flow-Motion [19] predicts the clean motion in conditional flow matching, MotionFlux [20] combines rectified flow with preference alignment, and MotionHiFlow [21] builds a hierarchical temporal flow with topology-aware encoding. Thus, flow matching itself is not our novelty: GALA fixes anatomical query-to-part identities, uses the resulting part tokens as DiT conditions, and supervises the flow endpoint with decoded-space kinematics. ST-GCN [22] and CTR-GCN [23] supply the topology prior. Part-level T2M [24, 25] generates or attends over limbs; GALA instead aligns language tokens to part latents inside one continuous flow. GENMO [13] is a generalist SMPL [26] model whose HumanML3D numbers pass through a conversion the authors call an inherent disadvantage; we list it as a reference, not as the target.

3. METHOD

Representation. A clip is $x_{1:T} \in \mathbb{R}^{T \times 263}$, $T \leq 196$, 20 fps, Guo features [1]: root motion, local joints, velocities, and four foot-contact channels. KIT-ML uses the analogous 251-D, 21-joint feature. Text is frozen CLIP ViT-B/32 [27] tokens $c_{1:N} \in \mathbb{R}^{N \times 512}$.

Part-aware graph tokenizer. Local joint positions and velocities form a J -node graph. Two CTR blocks mix a static anatomical adjacency with a sample-dependent attention graph; an ST-GCN variant (static adjacency only) and a graph-free conv VAE are the tokenizer controls. Joints are pooled into $P=5$ parts $\mathcal{P} = \{\text{torso, L-arm, R-arm, L-leg, R-leg}\}$, giving a global frame feature z_g and part features $z_p \in \mathbb{R}^{P \times d}$. A 1-D conv VAE [28] with stride 4 emits $\mu, \log \sigma^2 \in \mathbb{R}^{L \times 256}$, $L = \lceil T/4 \rceil$. Stage-1 training is

$$\mathcal{L}_{\text{VAE}} = \mathcal{L}_{\text{rec}} + \lambda_{\text{KL}} \mathcal{L}_{\text{KL}} + \lambda_{\text{vel}} \mathcal{L}_{\text{vel}} + \lambda_{\text{bone}} \mathcal{L}_{\text{bone}}. \quad (1)$$

After 35k AdamW steps the tokenizer is frozen.

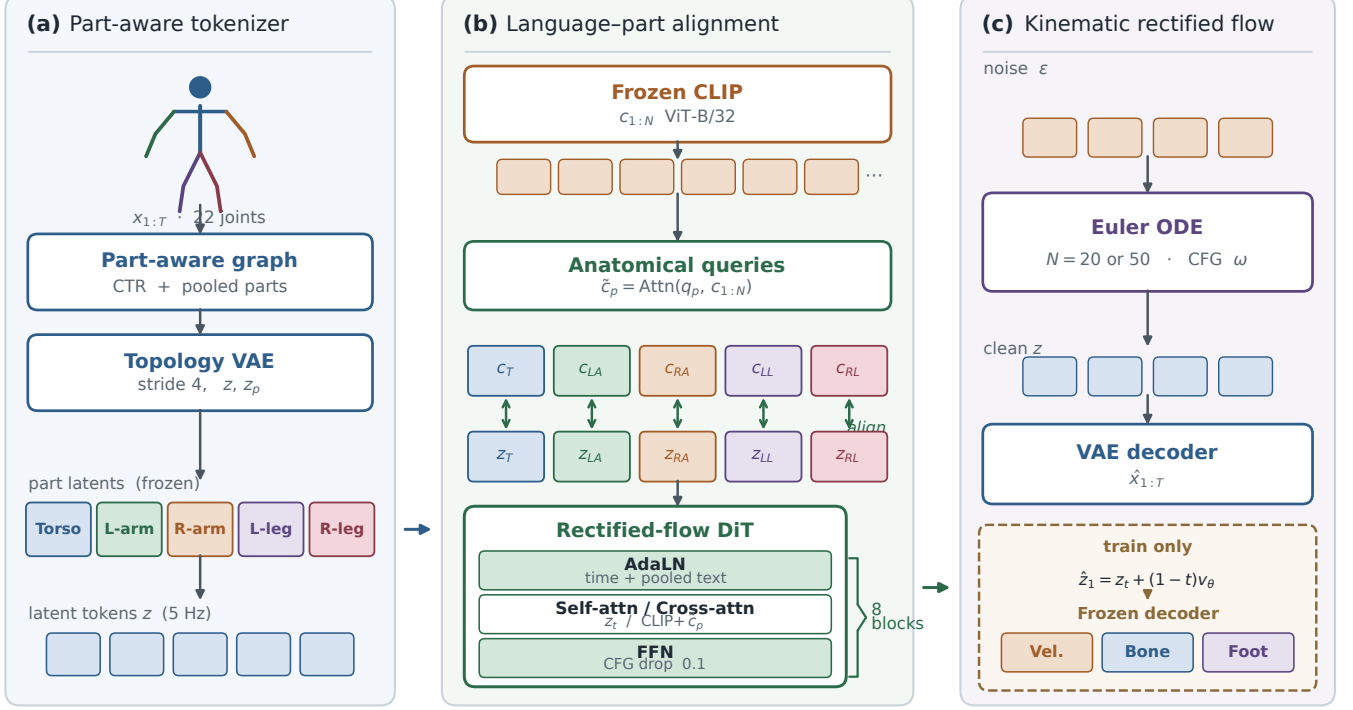


Fig. 1. GALA, three causally linked stages. **(a)** A skeleton-graph VAE emits a 5 Hz latent and five part codes (torso, L-arm, R-arm, L-leg, R-leg). **(b)** Learnable anatomical queries attend to CLIP tokens and align to the matching part latents; $\tilde{c}_p$ condition the DiT. **(c)** Inference is a 20- or 50-step Euler ODE. A training-only branch decodes $\hat{z}_1$ and penalizes velocity, bone length, and foot contact.

Fine-grained language-motion alignment. Global alignment remains symmetric InfoNCE on pooled CLIP and pooled latents, $\mathcal{L}_{\text{global}} = \text{InfoNCE}(\bar{c}, \bar{z})$, weight 0.25. On top of that, P learnable anatomical queries q_p attend to CLIP tokens,

$$\tilde{c}_p = \text{Attn}(q_p, c_{1:N}), \quad (2)$$

and are trained to match the corresponding part latent

$$\mathcal{L}_{\text{part}} = \frac{1}{P} \sum_{p=1}^P \text{SymNCE} \left(\{z_{b,p}\}_{b=1}^B, \{\tilde{c}_{b,p}\}_{b=1}^B \right), \quad (3)$$

where negatives are other examples in the minibatch. Each query is contrasted only with the graph-pooled latent at the same ordered index: torso, left arm, right arm, left leg, and right leg. This predefined correspondence prevents permutation ambiguity among the five slots. The same $\tilde{c}_p$ are concatenated to CLIP tokens as extra DiT context, so the branch is used at sampling time rather than only as a projection head. We do not equate attention weights with explanations: the diagnostic below establishes non-collapse, while the fixed-index contrastive targets provide anatomical supervision.

Kinematic rectified flow. Given clean μ (detached), $t \sim \mathcal{U}(0, 1)$, $\varepsilon \sim \mathcal{N}(0, I)$,

$$z_t = (1-t)\varepsilon + t\mu, \quad v^* = \mu - \varepsilon. \quad (4)$$

An 8-block DiT (width 512, 8 heads) predicts $v_\theta(z_t, t, c)$ with AdaLN modulation on the pooled text vector, self-attention on latent tokens, and cross-attention to CLIP+part tokens. Condition dropout 0.1 enables CFG [12]. The flow loss is $\mathcal{L}_{\text{RF}} = \|v_\theta - v^*\|_2^2$. We additionally

decode the endpoint estimate $\hat{z}_1 = z_t + (1-t)v_\theta$ with the frozen decoder and penalize motion-space kinematics:

$$\mathcal{L}_{\text{vel}} = \frac{1}{T-1} \sum_t \|(\hat{x}_{t+1} - \hat{x}_t) - (x_{t+1} - x_t)\|_1, \quad (5)$$

$$\mathcal{L}_{\text{bone}} = \sum_{(i,j) \in E} \|\hat{p}_i - \hat{p}_j\|_2 - \|p_i - p_j\|_2, \quad (6)$$

plus an ℓ_1 term $\mathcal{L}_{\text{foot}}$ on the four foot-contact channels. The generator objective is

$$\mathcal{L} = \mathcal{L}_{\text{RF}} + \lambda_g \mathcal{L}_{\text{global}} + \lambda_p \mathcal{L}_{\text{part}} + \lambda_v \mathcal{L}_{\text{vel}} + \lambda_b \mathcal{L}_{\text{bone}} + \lambda_f \mathcal{L}_{\text{foot}}. \quad (7)$$

Sampling is Euler, $z \leftarrow z + \frac{1}{N}(v_\theta + \omega(v_c - v_\theta))$, with $N \in \{10, 20, 50\}$ and $\omega \in \{2.0, 2.5\}$.

Contact-aware skating loss. The existing $\mathcal{L}_{\text{foot}}$ reconstructs the four GT foot-contact channels but does not penalize physical skating. We add a contact-aware skating loss on the decoded endpoint $\hat{X} = D(\hat{z}_1)$. World-frame foot positions $\hat{p}_{t,f}$ are recovered from the 263-D Guo features via `recover_ric`. The GT contact mask $c_{t,f}$ is taken from the last four binary channels (GT-contact-conditioned generated displacement):

$$\mathcal{L}_{\text{skate}} = \frac{\sum_{t,f} c_{t,f} \|\hat{p}_{t+1,f} - \hat{p}_{t,f}\|_2^2}{\sum_{t,f} c_{t,f} + \epsilon}. \quad (8)$$

Because $\mathcal{L}_{\text{skate}} \sim 10^{-4}$ while $\mathcal{L}_{\text{RF}} \sim 60$, a fixed weight would need $\lambda \sim 3 \times 10^5$. We instead use *adaptive weighting* $\lambda_{\text{eff}} = \eta \mathcal{L}_{\text{RF}} / \mathcal{L}_{\text{skate}}$

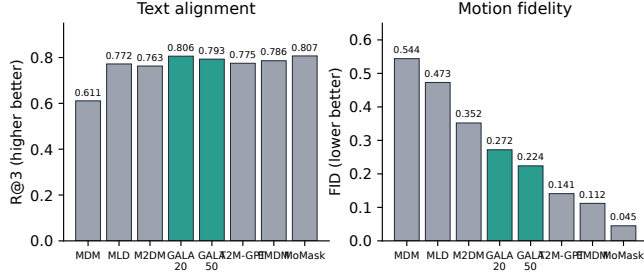


Fig. 2. R@3 and FID on HumanML3D. Teal is GALA. Discrete / few-step SOTA (MoMask, EMDM) still lead FID; GALA’s advantage is 20-NFE semantic accuracy.

so that $\lambda_{\text{eff}} \mathcal{L}_{\text{skate}} / \mathcal{L}_{\text{RF}} \approx \eta$ automatically. A decay schedule $\eta:0.03 \rightarrow 0.01$ over the first 40k steps recovers FID while retaining skating reduction.

Training. Stage 1: VAE only, 35k steps, effective batch 16, bf16. Stage 2: freeze graph+VAE, train flow+alignment, 100k steps, lr 2×10^{-4} , EMA 0.9999. The graph+global control in Table 1 uses the same tokenizer and DiT budget as the full model. Full GALA adds part alignment and decoded-space kinematic supervision.

4. EXPERIMENTS

Protocol. Official HumanML3D test [1], 4544 clips, Guo / MDM evaluator [2], batch 32, 20 replications, mean $\pm$ 95% CI. Diversity is compared to real motion (9.503), not maximized. KIT-ML uses the same evaluator on the 251-D official split. Hardware: RTX 4060 8 GB. GALA has 59.9M parameters (DiT 53.6M, VAE 3.2M, graph 1.6M).

Literature rows in Table 1 for T2M / MDM / M2DM / EMDM / GENMO follow GENMO Table 4 [13]; MLD / T2M-GPT / MoMask follow MoMask [8]. GENMO is listed as a generalist reference, not as the comparison target.

4.1. HumanML3D

Fig. 2 and Table 1 share one reading.

Semantic metrics at low NFE. At 20 NFEs GALA reaches R@3 $0.806 \pm .002$ and MM Dist $3.061 \pm .009$, improving the graph+global control by 0.038 and 0.187, respectively. It matches MoMask’s reported R@3 (0.807) within uncertainty while using a continuous 20-step trajectory. R@1 / R@2 are $0.494 \pm .003$ / $0.701 \pm .002$; Diversity 9.69 remains close to Real 9.50.

FID is the remaining gap, and it is a generator gap. GALA’s 50-NFE FID 0.224 improves the graph+global control (0.312) and M2DM (0.352), but trails T2M-GPT / EMDM / MoMask (0.141 / 0.112 / 0.045). We do not claim SOTA fidelity. Val reconstruction FID is 0.003: the Guo extractor barely distinguishes reconstructed clips from real ones, so the residual 0.224 generated FID is a generator rather than tokenizer bottleneck.

GENMO’s SMPL $\leftrightarrow$ HumanML3D conversion makes its FID 0.216 a weak target for a native-HML3D model. We report it for completeness and do not organize the paper around beating it.

4.2. KIT-ML

Table 2 uses the same Guo protocol on official KIT-ML. Literature numbers are from MoMask [8] and MDM [2]. GALA is trained with

the 21-joint graph and otherwise the same recipe as on HumanML3D.

4.3. What each module contributes

Reviewers need to know whether the graph and the alignment *do* anything. Table 3 is the component ablation: Base RF is a conv tokenizer with no InfoNCE; subsequent rows add the graph, global alignment, part alignment, and kinematic flow. Every row uses the same training budget, latent width, DiT, optimizer, and official 20-rep test protocol. Intermediate 320-clip validation runs select checkpoints but are never entered into this table.

The graph alone lowers FID by 0.033; part-indexed alignment recovers the retrieval loss introduced by global-only alignment and lowers FID by another 0.035; kinematic supervision yields the final 0.030 reduction. Before query-diversity regularization, the mean off-diagonal cosine between the five attention profiles was 0.9988; after repair it is 0.3474 over three fixed prompts. This is evidence against slot collapse, not a claim that raw attention is a faithful lexical explanation.

4.4. Physical constraints ablation

Table 4 isolates the contribution of the contact-aware skating loss. All variants continue training 50k steps from the same Distinct checkpoint with identical seed, hparams, NFE=20, CFG=2.5, and the official 20-replication test protocol. A3 is the $\lambda=0$ control; “Skate” uses fixed $\eta=0.02$; “Skate-decay” uses $\eta:0.03 \rightarrow 0.01$. We also ablate two TMM extensions: an anatomically anchored part alignment (Anchor) that regularises the five learnable queries toward cached CLIP embeddings of body-part prompts, and a spatio-temporal CTR graph (ST-CTR) that adds per-joint temporal self-attention to the spatial CTR block.

The skating loss reduces foot skating by 25% ($0.00155 \rightarrow 0.00117$) at fixed η , or 15% (0.00131) with the decay schedule, with CI non-overlapping in both cases. The decay schedule recovers most of the FID cost ($0.446 \rightarrow 0.403$) and R@3 ($0.794 \rightarrow 0.799$), making it the recommended configuration: **skating** -15% , **FID** $+0.049$, **R@3** -0.008 vs control. The anatomical anchor adds near-zero quality cost (FID $+0.015$, R@3 -0.002 , both within CI), confirming that the five queries can be grounded in CLIP semantics without hurting generation. ST-CTR adds temporal modelling but at a quality cost (FID $+0.104$); the temporal gate starts at zero and may need a warmer schedule. The full TMM combination accumulates costs rather than showing synergy, confirming that the physical benefit comes from the skating loss alone.

4.5. Tokenizer: graph vs. conv

Table 5 compares equally trained conv, static ST-GCN, and adaptive CTR tokenizers. Graph encoding cuts FID from 0.012 to 0.003 and MPJPE from 0.105 to 0.080, while R@3 is unchanged. ST-GCN already captures the FID gain; CTR’s distinct benefit is lower MPJPE ($0.088 \rightarrow 0.080$).

4.6. Sampling and the quality–efficiency Pareto

The val scan (Table 6) exposes a semantic–fidelity trade-off: 20 steps / CFG 2.5 favors R@3, whereas 50 / 2.0 favors FID. On an RTX 4060 (bf16, batch 1, $T=196$), 10/20/50 NFE take 70/140/340 ms (10 timed runs after warm-up), corresponding to 14.4/7.1/2.9 clips/s.

Table 1. HumanML3D test, Guo protocol, 20 replications on 4544 clips (mean $\pm$ 95% CI). $\uparrow$ higher, $\downarrow$ lower, $\rightarrow$ closer to Real.

Method	Rep.	NFE $\downarrow$	R@3 $\uparrow$	FID $\downarrow$	MM Dist $\downarrow$	Diversity $\rightarrow$
Real [1]	HML3D	–	0.797	0.002	2.974	9.503
T2M [1]	HML3D	–	0.740	1.067	3.340	9.188
MDM [2]	HML3D	1000	0.611	0.544	5.566	9.559
MLD [6]	HML3D	50	0.772	0.473	3.196	9.724
M2DM [9]	HML3D	–	0.763	0.352	3.134	9.926
T2M-GPT [7]	HML3D	–	0.775	0.141	3.121	9.761
EMDM [17]	HML3D	10	0.786	0.112	3.110	9.551
MoMask [8]	HML3D	–	0.807	0.045	2.958	–
GENMO [13]	SMPL	–	0.632	0.216	3.466	11.342
Graph+Global, CFG 2.5	HML3D	20	0.768 $\pm$.002	0.330 $\pm$.011	3.248 $\pm$.008	9.529 $\pm$.085
Graph+Global, CFG 2.0	HML3D	50	0.749 $\pm$.003	0.312 $\pm$.009	3.342 $\pm$.009	9.365 $\pm$.115
GALA, CFG 2.5	HML3D	20	0.806 $\pm$.002	0.272 $\pm$.007	3.061 $\pm$.009	9.692 $\pm$.087
GALA, CFG 2.0	HML3D	50	0.793 $\pm$.003	0.224 $\pm$.006	3.107 $\pm$.008	9.643 $\pm$.082

Table 2. KIT-ML test, Guo protocol; GALA uses 20 replications and 20 NFEs.

Method	R@3 $\uparrow$	FID $\downarrow$	MM Dist $\downarrow$	Diversity $\rightarrow$
Real	0.779	0.031	2.788	11.08
T2M [1]	0.681	3.022	3.488	–
MDM [2]	0.396	0.497	9.191	10.85
MLD [6]	0.734	0.404	3.204	–
T2M-GPT [7]	0.745	0.514	3.007	–
MoMask [8]	0.781	0.204	2.779	–
GALA	0.804 $\pm$.006	0.402 $\pm$.020	2.674 $\pm$.016	11.15 $\pm$.11

Table 3. Component ablation on HumanML3D under one official 20-rep test protocol (20 NFEs, CFG 2.5).

Model	Graph	Glob.	Part	Kin.	R@3 $\uparrow$	FID $\downarrow$
Base RF					0.789	0.402
+ Graph	✓				0.797	0.369
+ Global	✓	✓			0.769	0.337
+ Part	✓	✓	✓		0.801	0.302
GALA	✓	✓	✓	✓	0.806	0.272

5. CONCLUSION

GALA links graph-derived part latents, part-indexed language alignment, and a kinematically supervised rectified flow. Official 20-replication experiments show that the complete chain improves 20-NFE FID from 0.402 to 0.272 and R@3 from 0.789 to 0.806, while 50 NFEs reach FID 0.224. The tokenizer is near-lossless, so closing the remaining fidelity gap calls for a stronger transport model rather than a larger codebook.

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Table 4. Physical constraints ablation on HumanML3D test (20 replications, 20 NFEs, CFG 2.5, 50k steps from Distinct). Skating and paired acceleration are GT-contact-conditioned generated metrics.

Model	FID $\downarrow$	R@3 $\uparrow$	MM $\downarrow$	Div. $\rightarrow$	Skate $\downarrow$	Acc. $\downarrow$
A3 (control)	0.354 $\pm$.010	0.807 $\pm$.002	3.064	9.744	0.00155	0.00443
+ Skate	0.446 $\pm$.008	0.794 $\pm$.002	3.148	9.723	0.00117	0.00427
+ Skate-decay	0.403 $\pm$.010	0.799 $\pm$.002	3.112	9.740	0.00131	0.00434
+ Anchor	0.369 $\pm$.008	0.805 $\pm$.002	3.062	9.743	0.00155	0.00442
+ ST-CTR	0.458 $\pm$.010	0.792 $\pm$.002	3.125	9.379	0.00147	0.00439
+ All (TMM)	0.502 $\pm$.010	0.787 $\pm$.002	3.166	9.449	0.00129	0.00432

Table 5. Tokenizer reconstruction on HumanML3D val (1504 clips, no sampling).

Tokenizer	R@3 $\uparrow$	Recon FID $\downarrow$	MPJPE $\downarrow$	Bone $\downarrow$	Vel. $\downarrow$
Conv-VAE	0.755	0.012	0.105	0.061	0.039
ST-GCN-VAE	0.748	0.003	0.088	0.050	0.038
CTR-Graph-VAE	0.750	0.003	0.080	0.051	0.038

Table 6. Val sampling scan, 1504 clips, one seed. Official tests use 20/2.5 for semantics and 50/2.0 for fidelity.

Steps	CFG	FID $\downarrow$	R@3 $\uparrow$	MM Dist $\downarrow$
10	1.0	1.233	0.628	4.062
10	2.5	0.382	0.771	3.204
20	2.5	0.335	0.781	3.164
50	2.0	0.290	0.753	3.301
50	2.5	0.322	0.767	3.177

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